26900346 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCMAT300 - A FIRST COURSE IN COMPLEX ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **5** questions

Each question carries **2** marks

1. State the Extended Cauchy Integral Formula for the nth derivative of an analytic function. [U] / [CO3]
2. What is the parametrization of line segment joining z_1 and z_2 [K] / [CO1]
3. Find the Maclaurin series for $f(z) = zsinz$ [U] / [CO6]
4. State Laurent's theorem. [K] / [CO6]
5. What is the integral representation of the residue of a function f? [K] / [CO3, CO5]
6. What are the singular points, if any, of the function $\frac{\sin z^2}{(z^2+1)}$? [K] / [CO2]
7. If $z = e^{i\theta}$, express $\cos\theta$ in terms of z. [U] / [CO7]
8. Write $\sin 2\theta$ in terms of $z = e^{i\theta}$. [E] / [CO7]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. Define a simply connected domain. Also, evaluate $\int_C \frac{ze^z}{(z^2+9)^5}\,dz$, where C denotes any closed contour lying in the open disk $|z| < 1$. [U] / [CO2]

10. Show that use an antiderivative for every contour extending from a point z_1 to a point z_2, [A] / [CO3]
$\int_C z^n\,dz = \frac{1}{n+1}\left(z_2^{n+1} - z_1^{n+1}\right), \qquad (n = 0, 1, 2, \ldots).$

11. Assume that the complex series $\sum z_n$ is convergent. Show that [U] / [CO2]
$\lim_{n\to\infty} z_n = 0$

12. (i) Define Taylor series [C] / [CO6]
(ii) Show that the Maclaurin series representation of $\frac{1}{1-z}$ is , $\frac{1}{1-z}$ = $\sum_{n=0}^{\infty} z^n \quad (|z| < 1)$.

13. Show that the singular point of the following functions is a pole. Determine the order of that pole and the corresponding residue: [U] / [CO2]
(a) $\frac{1-coshz}{z^3}$ (b) $\frac{1-e^{2z}}{z^4}$.

14. Determine the singularities of $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ and evaluate the corresponding residues. [U] / [CO2]

15. Find P.V. $\int_{-\infty}^{\infty} \frac{x}{x^2+9}\,dx.$ [E] / [CO7]

16. Find $\int_0^{\infty} \frac{dx}{x^4+16}$. [E] / [CO7]

[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. State and prove the Maximum Modulus Principle [An] / [CO1, CO4]

18. Prove that if $z_n = x_n + iy_n$ $(n = 1, 2, \ldots)$ and $z = x + iy$, then [U] / [CO2]
$\lim_{n\to\infty} z_n = z$ if and only if $\lim_{n\to\infty} x_n = x \quad \text{and} \quad \lim_{n\to\infty} y_n = y.$

19. Expand $f(z) = \frac{-1}{(z-1)(z-2)}$ in powers of z and state the region of validity. [A] / [CO6]

20. *Let* $f(z) = z^2/z^2 - 1$ [A] / [CO5]

a) Find all the poles of $f(z)$ and determine their order

b) Find the residue of $f(z)$ *at each pole.*

21. Using residues, evaluate the improper integral $\int_{-\infty}^{\infty} \frac{x \sin ax}{x^2+4}\, dx \quad (a > 0)$ [E] / [CO7]

22. [A] / [CO7]

1. State and explain Jordan's Lemma.
2. Find the Cauchy principal value of the integral $\int_{-\infty}^{\infty} \frac{x \sin x}{x^2+2x+2}\, dx$

[10x3 = 30]